BIOLOGY FORM FOUR PRENATIONAL EXAMINATION 2024

MARKING SCHEME BIOLOGY 1

SECTION A (16 MARKS)

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	D	D	D	C	E	D	A	A	B

@ 1mark=10Marks

2.

i	Ii	iii	iv	v	vi
G	E	A	H	F	C

@ 1mark=6marks

SECTION B (54 MARKS)

3. Biology should not be taken as an optional subject at O'level because:

i)it helps us to understand ourselves better

ii)it helps us to understand and take care of our environment

iii)it helps us to understand and take care of other organisms

iv)it helps to improves variety of plants and animals

v)it helps us to get professional skills in medicine, nursing and pharmacy

vi)it helps us to understand the causes, symptoms, mode of transmission and treatment of diseases.

Any six correct points @1.5 marks Total =9 marks

4(a)Factors affecting the content of water and salts in the human body

i)weather -during cold weather there is no sweating, as a result the blood osmotic pressure decrease due to high rate of water retention following this the frequency of urination increases to remove any extra water from the body

ii)exercise -people who exercise will sweat and urinate more than those who do not exercise

iii)water intake-the more water we drink the more water we will have in our bodies

iv)failure of the pituitary gland to produce antidiuretic hormone, on the other hand causes kidney to cause shortage of water in the body.

Any three points @2marks=6marks

(b)i) Antidiuretic hormone

ii)Aldosterone hormone

@1.5 marks= (03marks)

Total= 9Marks

5(a)The way a plant cell prevents bursting when placed in hypertonic solution.

-When a plant cell is placed in hypertonic solution it absorbs water by osmosis, entering water creates turgor pressure and the cell wall creates wall pressure which when they reach at equilibrium the cell becomes turgid and does not burst. (03 marks)

(b)The effect in a recipient's blood when he has been transfused an incompatible blood from the donor.

-The recipient's blood undergo agglutination which lead to death. (03 marks)

(c)Precautions to be taken before blood transfusion

-the blood from the donor should be well screen on compatibility

-the donor's blood should be free from any infection

-blood transfusion should be done by a qualified person

-the transfused blood should be kept in a special bag and stored in a refrigerator for not more than 21 days

Any three points @ 1mark= (03 marks)

Total =9 marks

6.(a)Group A-insect pollinated flowers

Group B-wind pollinated flowers

@ 1.5mark=(0 3marks)

Flowers in group A(insect pollinated flowers)	Flowers in group B(Wind pollinated flowers)
i)they have brightly colored petals	-they have dull colored petals
ii)they produce nectar	-they don't produce nectar
iii)they produce scent	-they don't produce scent
iv)produce few pollen grains	-produce many pollen grains

@1.5 Marks=(06 marks)

Total =09 MARKS

7.(a)Mitosis,is a type of cell division that result to the formation of two diploid cells (03 marks)

(b)i) brings about growth

ii)bring about cell replacement

iii)bring about asexual reproduction

iv)brings about regeneration

@1.5 marks=(06marks)

Total =09 MARKS

8.(a)Mendel's first law of inheritance states that" the characteristics of an organism are controlled by internal factors (genes) which occur in a pair and only one pair is carried in a single gamete"(03 marks)

(b)

DEOXYRIBONUCLEIC ACID(DNA)	RIBONUCLEIC ACID(RNA)
i)it is double stranded	it is single stranded
ii)it contain deoxyribose sugar	it contains ribose sugar
iii)it is chemically stable	it is chemically unstable
iv)nitrogenous bases present are adenine,guanine,cytocine and thiamine	v)nitrogenous bases are adenine,guanine, cytocine and uracil

@01 mark =(04 marks)

(c)Bean seeds placed in a closed bottle containing cold water:

-The seeds did not germinate due to lack of oxygen for aerobic respiration and temperature for activation of enzymes (02 marks)

Total =09 MARKS eggs

SECTION C(30 MARKS)

9.Birth control refers to any method,medicine or device used to prevent pregnancy. (01 marks)

(a)Birth control methods that prevent union of gametes

i)calendar (rhythm)method,this involves partners to abstain from sexual intercourse during fertile days (02 marks)

ii)coitus interruption(withdrawal method),this involves a man to withdraw or remove penis from the vagina before ejaculation(02 marks)

iii)the use of condoms, condoms are put on the penis to trap sperms and avoid them from.

being deposited into the vagina (02 marks)

iv)sterilization,this involves cutting the Sperm duct in man(vasectomy) and on duct in woman (tubal ligation)to avoid gametes from coming into contact (02 marks)

(b) methods which suppress the formation or release of gametes

i)the use of implants,this involves implantation of a substance with hormones (estrogen and progesterone)hormones which prevent the follicles in ovaries from producing eggs (02 marks)

ii)contraceptive pills,the pills with oestrogen and progesterone hormones are taken daily to interfere with normal production of eggs .(02 marks)

(c)methods which prevent implantation of fertilized egg

i)the use of interuterine devices,this involves putting in special device into the uterus to avoid implantation of the zygote to the uterus (02 marks)

Total =15 Marks

10.Introduction:

skeleton is the rigid frame work of the body made up of bones that support the body (02 marks)

Importance of skeleton (Main body)

i)it helps to protect internal organs from mechanical damage eg.brain by the skull

ii)helps in production of red blood cells and storage of minerals eg.calcium

iii)it helps to give the body it's shape and support

iv) it provides surface area for attachment of muscles which facilitate movement

@03 marks=12 marks

Conclusion :any relevant conclusion (01 mark)

Total=15 MARKS

11.Introduction :

-Bacteria are single celled organisms that are found almost every where on th earth's surface (any relevant introduction) (01 Mark)

Main body.:

Reasons to why all bacteria should not be eliminated :

(importance of bacteria)

i)some bactria are used in making of antibiotics eg.penicillium

ii)bacteria are used in making biogas

iii)Some bactria are used in nitrogen fixation eg.those live in nodules of leguminous plants

iv)Some bacteria help in the production of vitamin K eg.those live in the large intestine of man

v)Some help in digestion of cellulose eg.those live in the appendix of herbivores

vi)Some are used in production of flavors in sour milk,cheese and yogurts

vii)they help in decomposition of dead organic matter to clean of our environment and ensure circulation of nutrients in the ecosystem .

Any 7 correct points @02 Marks

Total = 15 marks